SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 0614 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

I Variganji Sumanth Babu / 192524447, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V. Sumanth Babu

Annexure A

Short Answer Test

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blue, and 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

(5 Marks)

Answer:

Bayes' theorem is used because the observation that the ball is blue gives us new information about which box was selected. We need the probability of Box C after observing a blue ball, not merely the probability of selecting Box C before drawing.

Step 1: Find the probability of selecting each box.

Box B is twice as likely to be selected as either Box A or Box C. Assign relative weights 1, 2 and 1. Their total is 4, so P(A) = 1/4, P(B) = 2/4 = 1/2 and P(C) = 1/4.

Step 2: Find the probability of drawing a blue ball from each box.

Box A contains 4 blue balls out of 12, so P(Blue|A) = 4/12 = 1/3. Box B contains 7 blue balls out of 15, so P(Blue|B) = 7/15. Box C contains 2 blue balls out of 15, so P(Blue|C) = 2/15. These are conditional probabilities because each value assumes that a particular box has already been selected.

Step 3: Calculate the overall probability of drawing blue.

By the law of total probability, P(Blue) = P(Blue|A)P(A) + P(Blue|B)P(B) + P(Blue|C)P(C). Substituting the values gives (1/3)(1/4) + (7/15)(1/2) + (2/15)(1/4) = 1/12 + 7/30 + 1/30 = 7/20.

Step 4: Apply Bayes' theorem.

P(C|Blue) = [P(Blue|C)P(C)] / P(Blue) = [(2/15)(1/4)] / (7/20) = (1/30)(20/7) = 2/21.

The likelihoods provide a useful reasonableness check: P(Blue|A) = 33.33%, P(Blue|B) = 46.67%, and P(Blue|C) = 13.33%. A blue ball is therefore much less likely to come from Box C than from Box B. Bayes' theorem weights each likelihood by the chance of selecting that box, then divides by the total probability of observing blue. This normalization makes the posterior probabilities for A, B and C add to 1. The prior P(C) = 25% applies before the ball colour is known; P(C|Blue) = 9.52% is the updated probability after the evidence is observed.

Check: P(C and Blue) = (1/4)(2/15) = 1/30, and the total probability of a blue ball is 7/20. Dividing these gives the posterior, (1/30)/(7/20) = 2/21.

Final answer and interpretation: P(C|Blue) = 2/21 ≈ 0.0952 = 9.52%. Therefore, after observing a blue ball, there is approximately a 9.52% probability that Box C was selected. This is lower than Box C's original 25% selection probability because only 2 of its 15 balls are blue, while Box A and Box B have higher blue-ball proportions. The observation makes Boxes A and B more plausible sources of the blue ball.

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C? (5 Marks)

Answer:

This is a conditional-probability problem. The fact that the selected car is blue changes the likelihood that it came from each parking section. Bayes' theorem combines the initial chance of selecting a section with the chance of obtaining a blue car from that section.

Step 1: Find the prior probability of each section.

Section B is twice as likely to be chosen as Section A or Section C. The relative weights are therefore 1:2:1, giving P(A) = 1/4, P(B) = 1/2 and P(C) = 1/4.

Step 2: Calculate each section's blue-car probability.

Section A has 3 blue cars among 10 cars, so P(Blue|A) = 3/10. Section B has 6 blue cars among 15 cars, so P(Blue|B) = 6/15 = 2/5. Section C has 2 blue cars among 15 cars, so P(Blue|C) = 2/15.

Step 3: Calculate the overall probability of selecting a blue car.

P(Blue) = P(Blue|A)P(A) + P(Blue|B)P(B) + P(Blue|C)P(C). Substitution gives (3/10)(1/4) + (2/5)(1/2) + (2/15)(1/4) = 3/40 + 1/5 + 1/30 = 9/120 + 24/120 + 4/120 = 37/120.

Step 4: Find the posterior probability for Section C.

Bayes' theorem gives P(C|Blue) = [P(Blue|C)P(C)] / P(Blue). Thus P(C|Blue) = [(2/15)(1/4)]/(37/120) = (1/30)(120/37) = 4/37 ≈ 0.1081.

The posterior can be smaller than the prior even though Section C is a possible source. Before seeing the car colour, Section C has probability 25%; after observing blue, its probability is 10.81%. This change is reasonable because only 2/15 of Section C's cars are blue, compared with 3/10 in A and 6/15 in B. The law of total probability is important here: the denominator includes the blue-car probability contributed by every section, not only Section C. Ignoring either the section-selection rates or the colour proportions would produce a biased answer.

A joint-probability check gives P(A and Blue) = (1/4)(3/10) = 3/40, P(B and Blue) = (1/2)(2/5) = 1/5, and P(C and Blue) = (1/4)(2/15) = 1/30. These contributions sum to 37/120. Section C contributes only 1/30 of all possible selections, and dividing that contribution by the total blue-car probability gives (1/30)/(37/120) = 4/37. This is the same result obtained directly from Bayes' theorem. The denominator is essential because the blue car could have come from any of the three sections.

Final answer and interpretation: The probability that the blue car came from Section C is 4/37, or approximately 10.81%. Although Section C had a 25% prior selection probability, only 2 of its 15 cars are blue. Section B is more likely to produce a blue car and is also selected more often, so the observation of a blue car lowers the probability that Section C was the source. The posterior probability must reflect both section usage and car-colour composition.

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binary classification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based on the given information, determine the missing False Negatives (FN) value, and then calculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis. (5 Marks)

Answer:

Let “malignant” be the positive class and “benign” be the negative class. A true positive (TP) is a malignant case correctly classified as malignant; a true negative (TN) is a benign case correctly classified as benign. A false positive (FP) is a benign case incorrectly flagged as malignant, while a false negative (FN) is a malignant case incorrectly classified as benign.

Step 1: Calculate the missing false negatives.

There are 1,800 samples in each class, and 20% are reserved for testing. Assuming a class-wise (stratified) split, the test set contains 0.20 × 1,800 = 360 malignant samples. Of these, TP = 210 were detected correctly. Therefore, FN = 360 − TP = 360 − 210 = 150.

Step 2: Calculate the evaluation metrics from the supplied confusion-matrix values.

For these calculations, use TP = 210, TN = 190, FP = 30 and the inferred FN = 150. The total represented by these four entries is 210 + 190 + 30 + 150 = 580.

Accuracy = (TP + TN)/(TP + TN + FP + FN) = (210 + 190)/580 = 400/580 = 68.97%. Accuracy is the proportion of all represented test cases that were classified correctly.

Precision = TP/(TP + FP) = 210/(210 + 30) = 210/240 = 87.50%. Among cases the model labels malignant, 87.50% are truly malignant according to the supplied figures.

Sensitivity (Recall) = TP/(TP + FN) = 210/(210 + 150) = 210/360 = 58.33%. The model detects 58.33% of the malignant cases and misses the remaining 41.67%.

Specificity = TN/(TN + FP) = 190/(190 + 30) = 190/220 = 86.36%. This measures how well the model correctly identifies benign cases in the supplied confusion matrix.

Interpretation and data limitation: The precision is relatively high, but sensitivity is low for a breast-cancer detection task because many malignant cases are missed. Such false negatives are especially concerning in medical screening, so these results do not support using the model as a standalone diagnostic system; clinical validation and professional review would be required.

Precision and sensitivity answer different questions. Precision asks whether predicted malignant cases are truly malignant, while sensitivity asks how many actual malignant cases are found. A high precision cannot compensate for missing many cancer cases; the false-negative rate of 41.67% is a serious concern.

The question’s figures are internally inconsistent. A 20% test split from 1,800 benign samples implies 360 benign test cases, but TN + FP = 190 + 30 = 220. Therefore, the supplied counts do not describe a complete 720-case stratified test set. The metrics above use the given TN and FP values and the inferred FN; the dataset counts should be checked before reporting final performance.

4. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

(a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4
(b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2
(c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0
(d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1
(e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3
(f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6
(g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9
(h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2

i) Identify the epoch at which the model begins to overfit.
ii) Suggest two effective techniques to reduce overfitting in this model.
iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Answer:

i) Epoch at which overfitting begins

The validation loss reaches its minimum value of 2.0 at Epoch 3. At Epoch 4, validation loss rises from 2.0 to 2.1 even though training loss continues to fall from 2.2 to 2.0. Therefore, overfitting first becomes visible at Epoch 4, after the best validation result at Epoch 3.

Training loss measures error on examples used to update model parameters; validation loss measures error on separate examples. Falling training loss is desirable, but it does not prove that the model will perform well on new data.

The trend becomes clearer after Epoch 4. Training loss drops from 2.0 at Epoch 4 to 1.3 at Epoch 8, while validation loss climbs from 2.1 to 3.2. The growing difference means the model continues fitting the training data while its performance on the validation data worsens. The best generalization among the listed epochs is therefore at Epoch 3, when validation loss is lowest.

ii) Two effective techniques to reduce overfitting

1. Early stopping: Monitor validation loss after every epoch and stop training when it stops improving for a chosen number of epochs (patience). Here, the lowest validation loss occurs at Epoch 3, so training should not continue blindly through the later epochs. The final model can be restored to the checkpoint with the lowest validation loss.

2. Regularization, such as dropout or L2 regularization: Dropout temporarily disables a random subset of units during training, so the model cannot rely excessively on a small set of activations. L2 regularization adds a penalty for large weights to the loss function, encouraging simpler parameter values. Either method can reduce memorization; the best choice should be selected using validation performance.

iii) Effect on generalization

Overfitting occurs when a model learns noise or training-specific details instead of broad patterns. Training performance may remain strong while predictions on unseen images, people or operating conditions become less reliable. Validation loss should therefore guide model selection.

5. Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

(a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%
(b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%
(c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%
(d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%
(e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%
(f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%
(g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%
(h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68%

i) Identify the epoch at which the model starts overfitting.
ii) Explain why validation accuracy decreases despite training accuracy increasing.
iii) Suggest two methods to improve validation performance.

Answer:

i) Identify when overfitting starts

Validation accuracy increases from 58% at Epoch 1 to its maximum of 73% at Epoch 5. At Epoch 6, it falls to 72%, although training accuracy rises from 80% to 85%. This is the first clear sign that training performance is improving while performance on the held-out validation data is getting worse. Therefore, overfitting starts at Epoch 6; Epoch 5 is the best checkpoint among the listed epochs by validation accuracy.

ii) Why does validation accuracy fall while training accuracy increases?

During training, the model repeatedly adjusts its parameters to classify the training examples correctly. As epochs increase, it can become very good at recognizing patterns specific to those examples, including accidental correlations or noise. This causes training accuracy to keep increasing. The validation set contains examples not used for those parameter updates, so it tests whether the learned patterns transfer beyond the training data.

When the model begins to overfit, the learned decision boundary may become too tailored to the training set. It correctly classifies more training examples but makes more mistakes on validation examples. The gap in this question illustrates the pattern: training accuracy rises from 80% at Epoch 5 to 92% at Epoch 8, whereas validation accuracy decreases from 73% to 68%. More training is therefore not automatically better.

iii) Two methods to improve validation performance

1. Early stopping: Save the model whenever validation accuracy improves and stop when it declines consistently. Based on these observations, Epoch 5 is the best listed checkpoint. A small patience value can avoid reacting to a single noisy fluctuation, but training should not continue indefinitely after the validation trend turns down.

2. Regularization or data augmentation: Dropout and L2 regularization discourage the model from relying too strongly on training-specific features. For image data, data augmentation can also create realistic variations in the training examples, such as modest flips, crops or rotations when appropriate for the task. These methods encourage the model to learn more robust features and should be assessed using a validation set that remains separate from training.

Conclusion: Validation accuracy peaks at Epoch 5, making Epoch 5 the best checkpoint despite higher training accuracy in later epochs.

6. A machine learning model shows the following loss values during training: (5 Marks)

(a) Epoch 1: Training Loss = 3.0, Validation Loss = 2.7
(b) Epoch 2: Training Loss = 2.6, Validation Loss = 2.3
(c) Epoch 3: Training Loss = 2.3, Validation Loss = 2.1
(d) Epoch 4: Training Loss = 2.1, Validation Loss = 2.0
(e) Epoch 5: Training Loss = 1.9, Validation Loss = 2.2
(f) Epoch 6: Training Loss = 1.7, Validation Loss = 2.5
(g) Epoch 7: Training Loss = 1.6, Validation Loss = 2.8
(h) Epoch 8: Training Loss = 1.4, Validation Loss = 3.1

i) At which epoch does the model achieve optimal generalization?
ii) Identify signs of overfitting from the data.
iii) Recommend two regularization techniques and explain their impact.

Answer:

i) Epoch of optimal generalization

Optimal generalization occurs at Epoch 4 because validation loss is lowest at 2.0. Validation loss decreases from 2.7 at Epoch 1 to 2.0 at Epoch 4, indicating that the model is improving on data not used for parameter updates. After Epoch 4, validation loss increases, so later checkpoints perform worse on the validation set.

ii) Signs of overfitting

The main sign is that the two curves move in opposite directions after Epoch 4. Training loss falls from 1.9 at Epoch 5 to 1.4 at Epoch 8. At the same time, validation loss rises from 2.2 to 3.1. The difference between the two losses grows, suggesting that the model is becoming better at fitting the training data but less effective on unseen examples.

Validation loss first rises from 2.0 at Epoch 4 to 2.2 at Epoch 5. Thus, Epoch 5 is the first epoch at which overfitting is evident, while Epoch 4 is the preferred saved model. A high training score by itself cannot rule out overfitting because it measures performance on examples the model has already used during learning.

iii) Two regularization techniques and their impact

1. L2 regularization: Add a term proportional to the sum of squared model weights to the training objective. Large weights are penalized, discouraging an unnecessarily complex model and often making predictions less sensitive to small changes in input. The regularization strength must be tuned because excessive penalization can cause underfitting.

2. Dropout: During training, randomly deactivate some network units. This discourages the model from depending on one particular set of features or connections and can improve robustness. Dropout is disabled during ordinary inference, with activations handled according to the implementation's standard scaling rules.

Loss also reflects the confidence of predictions. A falling training loss with a rising validation loss means the model is becoming more accurate or confident on familiar examples while making worse predictions on held-out examples.

Early stopping is also useful in this example: monitor validation loss and retain the Epoch 4 checkpoint. These methods do not guarantee better performance automatically, so their impact should be checked on validation data and the final model should be evaluated on a separate test set.

7. In an industrial process, an AI robot is assigned to segregate the front and back tires automatically. Let the number of tires are 250 (125 front and 125 back tires).

(5 Marks)

a) Front identified as Front: 119
b) Front identifies as Back: 06
c) Back identified as Back: 120
d) Back identified as Front: 05

Compute: Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Answer:

Step 1: Construct the confusion-matrix counts.

Treat “Front” as the positive class. Front tyres correctly identified as Front are true positives: TP = 119. Front tyres incorrectly identified as Back are false negatives: FN = 6. Back tyres correctly identified as Back are true negatives: TN = 120. Back tyres incorrectly identified as Front are false positives: FP = 5. The total is 119 + 6 + 120 + 5 = 250, which matches the stated number of tyres.

Step 2: Calculate accuracy.

Accuracy = (TP + TN)/(TP + TN + FP + FN) = (119 + 120)/250 = 239/250 = 0.956 = 95.60%. Thus, 239 of the 250 tyres are classified correctly, while 11 are classified incorrectly.

Step 3: Calculate precision.

Precision = TP/(TP + FP) = 119/(119 + 5) = 119/124 = 0.9597 = 95.97%. When the robot predicts “Front”, about 95.97% of those predictions are actually front tyres. The five false positives represent back tyres that would be sent to the front-tyre group.

Step 4: Calculate sensitivity (recall).

Sensitivity = TP/(TP + FN) = 119/(119 + 6) = 119/125 = 0.952 = 95.20%. The robot correctly recognizes 95.20% of all front tyres. The six false negatives are front tyres that the robot mistakenly labels Back.

Step 5: Calculate specificity.

Specificity = TN/(TN + FP) = 120/(120 + 5) = 120/125 = 0.96 = 96.00%. This means 96% of back tyres are correctly identified as Back.

Step 6: Calculate the F1-score.

F1 = 2TP/(2TP + FP + FN) = 2(119)/(238 + 5 + 6) = 238/249 = 0.9558 = 95.58%. The F1-score is the harmonic mean of precision and recall, so it is high only when both are high.

The counts are consistent: TP + FN = 125 front tyres and TN + FP = 125 back tyres. Precision and recall are close, which is why the F1-score is also close to both. Reporting all metrics is more informative than accuracy alone because the two kinds of misclassification can have different operational costs.

Final interpretation: All five metrics are around 95% or higher, indicating strong performance on this sample. However, even a small error rate can matter on an automated production line: a mislabelled tyre may cause a later assembly error or require rework. The system should therefore be monitored on new batches, and the cost of false positives and false negatives should be considered before deployment.

8. In a medical diagnosis system, an AI model detects whether a patient has a disease or not. Out of 300 cases (150 diseased and 150 healthy): (5 Marks)

a) Diseased identified as Diseased: 138
b) Diseased identified as Healthy: 12
c) Healthy identified as Healthy: 135
d) Healthy identified as Diseased: 15

Compute: **Accuracy, Precision, Sensitivity (Recall), Specificity, and F1-Score.**

Answer:

Step 1: Identify the confusion-matrix values.

Treat “Diseased” as the positive class. The counts are TP = 138, FN = 12, TN = 135 and FP = 15. The total number of cases is 138 + 12 + 135 + 15 = 300, agreeing with the question. TP represents diseased patients detected correctly; FN represents diseased patients the model misses; TN represents healthy patients correctly identified; and FP represents healthy patients incorrectly flagged as diseased.

Step 2: Accuracy.

Accuracy = (TP + TN)/Total = (138 + 135)/300 = 273/300 = 91.00%. The classifier makes the correct decision for 273 out of 300 cases.

Step 3: Precision.

Precision = TP/(TP + FP) = 138/(138 + 15) = 138/153 ≈ 90.20%. When the model predicts that a patient is diseased, approximately 90.20% of those positive predictions are correct. The remaining positive predictions are false alarms among healthy patients.

Step 4: Sensitivity (recall).

Sensitivity = TP/(TP + FN) = 138/(138 + 12) = 138/150 = 92.00%. The model detects 92% of all diseased patients in the test sample. It misses 12 of the 150 diseased cases, so the false-negative rate is 8%.

Step 5: Specificity.

Specificity = TN/(TN + FP) = 135/(135 + 15) = 135/150 = 90.00%. The system correctly recognizes 90% of healthy patients. The other 10% are incorrectly flagged for possible disease.

Step 6: F1-score.

F1 = 2TP/(2TP + FP + FN) = 276/(276 + 15 + 12) = 276/303 ≈ 91.09%. This combines precision and recall into one score and is useful when both missed cases and false alarms matter.

Final interpretation: The model shows reasonably balanced performance, with 91% accuracy, 92% sensitivity and 90% specificity. In medical screening, sensitivity is especially important because a false negative may delay further examination. However, the 12 missed cases show why the model must not be treated as infallible. These metrics describe only the given dataset; clinical use would require independent validation, suitable thresholds and qualified medical oversight.

The 8% missed-case rate and 10% false-alarm rate should be reviewed with clinical experts and confirmed on an independent test set before any real-world use.

9. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam): (5 Marks)

- Spam identified as Spam: 180
- Spam identified as Not Spam: 20
- Not Spam identified as Not Spam: 170
- Not Spam identified as Spam: 30

Compute: **Accuracy, Precision, Sensitivity, Specificity, and F1-Score.**

Answer:

Step 1: Identify the confusion-matrix counts.

Treat “Spam” as the positive class. The data gives TP = 180 spam emails correctly identified as spam, FN = 20 spam emails incorrectly labelled Not Spam, TN = 170 non-spam emails correctly retained as Not Spam, and FP = 30 legitimate emails incorrectly labelled Spam. The sum is 180 + 20 + 170 + 30 = 400 emails.

Step 2: Accuracy.

Accuracy = (TP + TN)/Total = (180 + 170)/400 = 350/400 = 87.50%. The system assigns the correct label to 350 emails.

Step 3: Precision.

Precision = TP/(TP + FP) = 180/(180 + 30) = 180/210 ≈ 85.71%. Of the emails placed in the spam folder, about 85.71% are actually spam. Lower precision means more legitimate emails are incorrectly diverted into spam.

Step 4: Sensitivity (recall).

Sensitivity = TP/(TP + FN) = 180/(180 + 20) = 180/200 = 90.00%. The filter catches 90% of all spam messages, but it misses 20 of the 200 spam emails. Those false negatives may reach the user’s inbox.

Step 5: Specificity.

Specificity = TN/(TN + FP) = 170/(170 + 30) = 170/200 = 85.00%. The filter correctly identifies 85% of non-spam emails. The remaining 15% of legitimate messages are incorrectly marked as spam.

Step 6: F1-score.

F1 = 2TP/(2TP + FP + FN) = 360/(360 + 30 + 20) = 360/410 ≈ 87.80%. The F1-score balances the precision and recall of the spam class.

The missed-spam rate is 20/200 = 10%, while the false-alarm rate is 30/200 = 15%. Thus, some spam reaches the inbox, and some legitimate mail is hidden in the spam folder.

Final interpretation: The filter catches most spam, as shown by 90% recall, but it also incorrectly flags 30 legitimate emails. In a practical system, false positives can be costly when they hide important messages, while false negatives allow unwanted or potentially harmful emails to remain in the inbox. The appropriate decision threshold depends on which error is more costly. The metrics should be checked periodically because email patterns and spam tactics change over time.

10. In a quality inspection system in manufacturing, an AI model identifies defective and non-defective products. Out of 500 products (250 defective and 250 non-defective): (5 Marks)

- Defective identified as Defective: 2 30
- Defective identified as Non-Defective: 20
- Non-Defective identified as Non-Defective: 225
- Non-Defective identified as Defective: 25

Compute: **Accuracy, Precision, Sensitivity, Specificity, and F1-Score.**

Answer:

Step 1: Interpret the supplied values and define the classes.

Treat “Defective” as the positive class. The question prints “Defective identified as Defective: 2 30”. This is interpreted as TP = 230 because there are 250 defective products and 20 are identified as non-defective, giving FN = 20 and TP = 250 − 20 = 230. The other values are TN = 225 and FP = 25. This interpretation makes the counts total 230 + 20 + 225 + 25 = 500, as stated.

Step 2: Accuracy.

Accuracy = (TP + TN)/Total = (230 + 225)/500 = 455/500 = 91.00%. The system classifies 455 of the 500 products correctly.

Step 3: Precision.

Precision = TP/(TP + FP) = 230/(230 + 25) = 230/255 ≈ 90.20%. Among products classified as defective, about 90.20% are actually defective. The 25 false positives are good products that the system unnecessarily sends for rejection or reinspection.

Step 4: Sensitivity (recall).

Sensitivity = TP/(TP + FN) = 230/(230 + 20) = 230/250 = 92.00%. The system detects 92% of all defective products. It misses 20 defective products, which may proceed to packaging or customers if no further inspection is performed.

Step 5: Specificity.

Specificity = TN/(TN + FP) = 225/(225 + 25) = 225/250 = 90.00%. The system correctly identifies 90% of non-defective products and falsely flags the remaining 10%.

Step 6: F1-score.

F1 = 2TP/(2TP + FP + FN) = 460/(460 + 25 + 20) = 460/505 ≈ 91.09%. It provides a single measure that balances the precision and recall of the defective class.

The false-negative rate is 20/250 = 8%, and the false-positive rate is 25/250 = 10%. Both escaped defects and unnecessary reinspection matter.

Final interpretation: The metrics indicate good but imperfect performance. Sensitivity matters because missed defects can affect product safety, reliability and customer satisfaction. Precision also matters because false alarms create scrap, reinspection and production delays. The value 230 is an explicit assumption based on the stated class total and false-negative count; the ambiguous printed value should be confirmed with the instructor or original question sheet before final submission.

11. In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated. (5 Marks)

Answer:

The problem does not explicitly name which shop's posterior probability is required. Since an adulterated bottle has been observed, calculate the probability that it came from each shop. Bayes' theorem updates the prior shop-selection probabilities using the adulteration rates of the two shops.

Step 1: Prior probabilities.

Shop Y is twice as likely to be selected as Shop X. Therefore, P(X) = 1/3 and P(Y) = 2/3.

Step 2: Probability of adulteration within each shop.

Shop X has 30 adulterated bottles out of 20 + 30 = 50, so P(Adulterated|X) = 30/50 = 3/5. Shop Y has 20 adulterated bottles out of 40 + 20 = 60, so P(Adulterated|Y) = 20/60 = 1/3.

Step 3: Find the overall probability of drawing an adulterated bottle.

P(Adulterated) = P(Adulterated|X)P(X) + P(Adulterated|Y)P(Y) = (3/5)(1/3) + (1/3)(2/3) = 1/5 + 2/9 = 9/45 + 10/45 = 19/45.

Step 4: Calculate the probability that the bottle came from Shop Y.

P(Y|Adulterated) = [P(Adulterated|Y)P(Y)]/P(Adulterated) = [(1/3)(2/3)]/(19/45) = (2/9)(45/19) = 10/19 ≈ 52.63%.

Step 5: Calculate the probability that it came from Shop X.

P(X|Adulterated) = [(3/5)(1/3)]/(19/45) = (1/5)(45/19) = 9/19 ≈ 47.37%. The two posterior probabilities add to 1, which is a useful check: 10/19 + 9/19 = 1.

Before the bottle is examined, the chance of choosing Y is 66.67%, but that is not the answer to the question after adulteration is observed. The observation changes the probability because the shops have different adulteration rates. The posterior probability considers only the cases compatible with the evidence and asks what fraction of those cases came from each source. This is the central distinction between a prior probability and a conditional probability.

A check using expected counts gives the same result. One third of selections come from X, where the adulteration rate is 60%, contributing 1/5 of all selections as adulterated X bottles. Two thirds come from Y, where the adulteration rate is 33.33%, contributing 2/9 as adulterated Y bottles. The total adulterated probability is 19/45, and the Y contribution is 2/9; dividing the contribution by the total gives (2/9)/(19/45) = 10/19. This contribution-based explanation is another way to see the logic of Bayes' theorem.

Final interpretation: Even though Shop Y is selected more often, it has a lower adulteration rate than Shop X. Nevertheless, because twice as many selections come from Shop Y, an adulterated bottle is slightly more likely to have come from Shop Y in this particular setup: 52.63% versus 47.37%. Bayes' theorem accounts for both the shop-selection likelihood and each shop's adulteration rate; looking at either factor alone would give an incomplete answer.

12. A warehouse has two sections. Section A contains 15 defective items and 35 non-defective items, while Section B contains 25 defective items and 25 non-defective items. One section is chosen at random, but Section A is three times as likely to be chosen as Section B. An item selected from the chosen section is found to be defective. (5 Marks)

Answer:

The question states that a defective item is observed but does not explicitly ask for a particular section's probability. Therefore, calculate both P(A|Defective) and P(B|Defective), so that the result fully answers the conditional-probability problem.

Step 1: Find the probabilities of selecting the sections.

Section A is three times as likely to be selected as Section B. The relative weights are 3:1, so P(A) = 3/4 and P(B) = 1/4.

Step 2: Find each section's defective-item rate.

Section A has 15 defective and 35 non-defective items, for a total of 50. Hence, P(Defective|A) = 15/50 = 3/10. Section B has 25 defective and 25 non-defective items, also 50 in total, so P(Defective|B) = 25/50 = 1/2.

Step 3: Calculate the overall probability of selecting a defective item.

P(Defective) = P(Defective|A)P(A) + P(Defective|B)P(B) = (3/10)(3/4) + (1/2)(1/4) = 9/40 + 5/40 = 14/40 = 7/20.

Step 4: Calculate the posterior probability for Section A.

P(A|Defective) = [P(Defective|A)P(A)]/P(Defective) = (9/40)/(14/40) = 9/14 ≈ 64.29%.

Step 5: Calculate the posterior probability for Section B.

P(B|Defective) = [P(Defective|B)P(B)]/P(Defective) = (5/40)/(14/40) = 5/14 ≈ 35.71%. These probabilities add up to 100%, because the chosen item must have come from either A or B.

The prior probability of A is 75%, but this is not the required conditional probability. Once a defective item is observed, the evidence must be considered. Section B has the higher defect rate, yet it is selected only one quarter of the time. The final answer therefore depends on both how often each section is selected and how likely that section is to produce a defective item. This is why the posterior for A, rather than merely its prior, must be calculated.

A useful check is to compare the joint contributions to the defective outcome. The probability of selecting A and then drawing a defective item is (3/4)(3/10) = 9/40. The probability of selecting B and drawing a defective item is (1/4)(1/2) = 5/40. Since the first contribution is larger, A has the larger posterior probability among defective items, even though its within-section defect rate is lower. Dividing each contribution by their sum, 14/40, produces probabilities that sum to 1 and gives the required updated belief after observing a defect.

Final interpretation: Given that an item is defective, the probability it came from Section A is approximately 64.29%, compared with 35.71% for Section B. Section A is selected three times as often, which offsets its lower defective-item rate. Section B has the higher defect rate (50% rather than 30%), but it is chosen less often. The posterior result combines both facts instead of assuming that the section with the highest defect rate must be the source.

13. In a binary classification problem using a deep learning model with a K-Nearest Neighbor classifier, the model produces True Positive (TP) = 120 and True Negative (TN) = 150.The dataset contains 1000 samples per class, and 20% of the data is used for testing. (5 Marks)
Find the following: Accuracy, Precision, Sensitivity (Recall), Specificity.

Answer:

Assumption: the 20% test split is stratified, so 20% of each class is included in the test set. Each class contains 1,000 samples, giving 200 positive and 200 negative test examples, or 400 test examples in total.

Step 1: Complete the confusion matrix.

TP = 120, so FN = 200 − 120 = 80. TN = 150, so FP = 200 − 150 = 50. Therefore, TP = 120, FN = 80, TN = 150 and FP = 50. This is consistent with 200 examples in each actual class: TP + FN = 200 positive examples, and TN + FP = 200 negative examples.

Step 2: Accuracy.

Accuracy = (TP + TN)/(TP + TN + FP + FN) = (120 + 150)/400 = 270/400 = 67.50%. It measures the fraction of all test samples classified correctly.

Step 3: Precision.

Precision = TP/(TP + FP) = 120/(120 + 50) = 120/170 ≈ 70.59%. Of the samples predicted as positive, about 70.59% are truly positive. The rest are false-positive predictions.

Step 4: Sensitivity (recall).

Sensitivity = TP/(TP + FN) = 120/(120 + 80) = 120/200 = 60.00%. The classifier finds 60% of the actual positive cases and misses 40% of them.

Step 5: Specificity.

Specificity = TN/(TN + FP) = 150/(150 + 50) = 150/200 = 75.00%. It correctly identifies three-quarters of the negative cases; the remaining 25% are false positives.

These measures are complementary: precision concerns the quality of positive predictions, sensitivity concerns the coverage of actual positives, and specificity concerns recognition of negatives. None should be interpreted in isolation. A model-selection decision should be based on the metric that reflects the real application's risk, and on validation results that are not reused for final testing.

The counts can be checked by class: the model correctly identifies 120 of 200 positive examples and misses 80; it correctly identifies 150 of 200 negative examples and incorrectly marks 50 as positive. Because sensitivity is only 60%, two out of every five positive test cases are missed in this sample. If positive cases represent a serious condition or costly failure, that level of recall may be insufficient even though specificity is 75%. Whether to adjust the decision threshold depends on the consequences of false negatives compared with false positives.

Final interpretation: The classifier has 67.50% accuracy, 70.59% precision, 60% sensitivity and 75% specificity under the stated stratified-split assumption. Specificity is higher than sensitivity, so the model is better at rejecting negative cases than detecting positive cases. Since 80 of 200 positive cases are missed, its recall may be inadequate where missing a positive case is costly. The figures describe the test split only; further tuning should be guided by validation data and the cost of different error types.

14. A medical image classification system using a deep learning framework combined with K-Nearest Neighbor (KNN) yields TP = 180 and TN = 160. Each class contains 1200 samples, and 25% of the dataset is used as test data. Calculate: Accuracy Precision Sensitivity Specificity. (5 Marks)

Answer:

Assumption: the test split contains 25% of each class. With 1,200 samples per class, there are 0.25 × 1,200 = 300 positive test examples and 300 negative test examples, making 600 test examples in total.

Step 1: Complete the confusion matrix.

TP = 180, so FN = 300 − 180 = 120. TN = 160, so FP = 300 − 160 = 140. The resulting counts are TP = 180, FN = 120, TN = 160 and FP = 140. Check: TP + FN = 300 positives and TN + FP = 300 negatives.

Step 2: Accuracy.

Accuracy = (TP + TN)/Total = (180 + 160)/600 = 340/600 ≈ 56.67%. This is only slightly above 50%, so accuracy alone does not show strong classification performance.

Step 3: Precision.

Precision = TP/(TP + FP) = 180/(180 + 140) = 180/320 = 56.25%. Of all cases predicted as positive, just over half are truly positive. The large number of false positives lowers precision.

Step 4: Sensitivity (recall).

Sensitivity = TP/(TP + FN) = 180/(180 + 120) = 180/300 = 60.00%. The model detects 60% of positive cases and misses 40%, or 120 out of 300.

Step 5: Specificity.

Specificity = TN/(TN + FP) = 160/(160 + 140) = 160/300 ≈ 53.33%. The model correctly identifies only about 53.33% of negative cases, while 46.67% of negatives are incorrectly labelled positive.

The confusion-matrix totals are internally consistent: 180 + 120 = 300 actual positive cases and 160 + 140 = 300 actual negative cases. Sensitivity and specificity are both relatively weak, with sensitivity at 60% and specificity at 53.33%. Thus, the model misses 120 positive cases and misclassifies 140 negative cases. The results suggest broad classification difficulty rather than just one isolated type of error. Feature processing, labels, the model configuration and the decision threshold should be examined using validation data before retraining.

The test set is balanced, with 300 examples in each actual class, so the weak accuracy is not caused by class imbalance. The false-negative rate is 120/300 = 40%, and the false-positive rate is 140/300 = 46.67%, confirming errors in both directions.

Final interpretation: The relatively low accuracy, precision and specificity suggest that the classifier is not performing well on this test set. It produces many false positives and still misses a substantial share of positive cases. Its decision threshold, feature quality, class labels and train/test separation should be reviewed. Any changes should be compared on validation data, and the final metrics should be measured once on a separate test set that was not used to select the model.

15. A deep learning model is trained for a classification task, and the following loss values are observed: (5 Marks)

i. **Epoch 1: Training Loss = 2.8, Validation Loss = 2.6**
ii. **Epoch 2: Training Loss = 2.4, Validation Loss = 2.2**
iii. **Epoch 3: Training Loss = 2.1, Validation Loss = 1.9**
iv. **Epoch 4: Training Loss = 1.9, Validation Loss = 1.8**
v. **Epoch 5: Training Loss = 1.7, Validation Loss = 1.9**
vi. **Epoch 6: Training Loss = 1.5, Validation Loss = 2.1**
vii. **Epoch 7: Training Loss = 1.3, Validation Loss = 2.4**
viii. **Epoch 8: Training Loss = 1.2, Validation Loss = 2.7**

Answer the following:

a. At which epoch does the model achieve the best generalization performance?

b. At which point does overfitting begin? Justify your answer.

c. Suggest two techniques to reduce overfitting and briefly explain how they help.

Answer:

a) Epoch of best generalization performance

The model achieves its best generalization performance at Epoch 4, where validation loss is at its minimum value of 1.8. Up to this point, both training loss and validation loss decrease: training loss falls from 2.8 at Epoch 1 to 1.9 at Epoch 4, and validation loss falls from 2.6 to 1.8. This indicates that learning is improving performance on both the training data and the held-out validation data.

b) Point at which overfitting begins

Overfitting first becomes visible at Epoch 5. Validation loss rises from 1.8 at Epoch 4 to 1.9 at Epoch 5, while training loss continues to decrease from 1.9 to 1.7. The trend continues in later epochs: by Epoch 8, training loss reaches 1.2 but validation loss has risen to 2.7. The model is becoming increasingly good at fitting training examples while becoming worse at predicting validation examples.

This divergence is the key evidence, rather than the training loss alone. The best checkpoint is Epoch 4 because it has the lowest validation loss. If training is continued through Epoch 8, the much lower training loss may look encouraging, but it does not represent improved generalization. The validation curve shows that the model is likely learning details, noise or accidental correlations that do not transfer well to unseen cases.

c) Two techniques to reduce overfitting

1. Early stopping: Evaluate validation loss at the end of each epoch, save the best checkpoint, and stop training when validation loss has failed to improve for a chosen patience interval. For the values shown, retain the Epoch 4 checkpoint. This prevents later epochs from replacing a model that generalizes better with one that only fits the training set more closely.

2. Dropout or L2 regularization: Dropout randomly disables some units during training, reducing dependence on particular activations. L2 regularization penalizes large weights and discourages an unnecessarily complex model. Tune either method using validation data.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand